AI + IoT in Automobiles

Proof Of Concept (PoC)

Kunal Bahuguna

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Background and Context

- A major automobile manufacturer based out of US, looking to digitally modernize their technology operations to provide adavanced AI driven analytics to proactively increase customer satisfaction, and brand reputation resulting in higher repeat customer sales and new sales of their cars.

Opportunity Analysis

- **Problem Statement**

  - Our automobile company faces challenges with our customers having to bear high repair costs due to their lack of their timely auto maintenance and sudden disruption to their daily lives average repair bills costing **$500-$1000 (*per AAA*)** for such breakdowns and incidents.

  - Auto insurers at the same time are unable to determine accurate premiums for a driver based on their driving record. This disconnect contributes to higher accident rates, fraudulent claims and suboptimal capital allocation for insurers and auto manufacturers. Change in driving behavior can result in reduction of accidents by **20%-35%** and hence lower claim payouts.

- **Pain Points**

  - **Drivers and auto owners:** Face unpredictable costs (**per AAA average repair bill ~ $500-$1000)** for each such incident

  - **Car Manufacturers:** Struggle with monitoring of sold vehicles performance, leading to reactive customer service and potential brand damage from accelerated mechanical failures of their cars.

  - **Insurers/Insurance companies:** Rely on inaccurate risk profiling of drivers, resulting in lower premiums for riskier driver and significant loss for insurers by lower underwriting resulting in millions of dollars of lost revenue each year.

  - **Societal, Civic & Regulatory impact:** Increased road accidents, primarily due to poor maintenance of vehicles, environmental wastes emitted from inefficient vehicles and their usage and regulatory pressures from government agencies due to inaccurate emission tracking.

Proposed Solution

- An Integrated AI + IoT platform that leverages vehicle telematics for predictive maintenance, seamless manufacturer notifications and behavior-based insurance incentives.

- This solution will collect vehicle data and and transform them into actionable insights, promoting active interventions, cost savings and safer roads.

INPUT:
- Vehicles telematics data
- Hardware Components
- Data Communication Infrastructure
- Cloud or Edge Platform
- Integration Interfaces
- Security & Access Control

AI Model

OUTPUT:
- Predictive Maintenance Alerts
- Manufacturer Notifications
- Behavior-Based Insurance Reports
- Vehicle Health Dashboards
- Cost & Safety Metrics
- Testing and Validation Data

Pros of the Proposed AI Solution

Theme	Description	Benefits
Operational Efficiency	• Real-time IoT data and AI predictions reduce unexpected breakdowns by forecasting issues.	• Lowers repair costs for auto owners
Safety Enhancements	• AI analyzes driving behaviors (e. g. braking patterns) to incentivize safe habits, while maintenance alerts prevent accidents like brake failures.	• Reduction in road accidents linked to poor maintenance
Financial Incentives and Savings	• Usage based insurance (UBI) rewards good drivers with discounts ranging from 10%-40% on their premiums • Insurers benefit from accurate risk profiling for drivers and fraud detection via verifiable IoT data logs.	• The company will save Millions of $$$'s annually on claims reimbursement • Drivers gain discounted pricing
Data-Driven Insights	• Aggregated data from customers will help in identifying common parts failures and allow us to fix the defective and faulty parts accordingly. • Proactively allow to improve quality on the defective parts, and/or will allow us to provide feedback to and negotiate with the parts vendors. • For Insurers: It allows them to offer better customized premium and incentives based on real driving data of the insured drivers and their and risk profiles.	• Enhanced product quality • Support environmental goals by optimizing fuel efficiency and reduce waste. • For Insurers: Less allocation of capital and better money management for claims & reimbursements.
Innovation and Scalability	• Will allow for further enhancement of the autonomous driving capabilities through building a data foundation and advancing the autonomous vehicle technologies such as Vehicle-to-Device (V2D) and Cellular Vehicle To Everything (CV2X): Vehicle to Vehicle (V2V), Vehicle to Infrastructure (V2I), Vehicle to Pedestrians (V2P) and Vehicle to Network (V2N)	• Accelerate the development of Autonomous vehicle technology development for us. • Build and realize the Vehicle to Everything (Cv2X) tech.

Cons of the Proposed AI Solution

Theme	Description	Impact
Privacy and Data Security	• Personal user data collection on a continuous basis such as locations and habits raises concerns over invasion of privacy and personal information misuse.	• Potential breaches could erode trust and non-compliance with General Data Protection Regulation (GDPR) compliances.
Technical and Reliability Issues	• Dependency on connectivity could delay timely alerts from the IoT devices • AI false positives e.g. misinterpreting sensor noise.	• Systems downtime risks • Vulnerability to cyber attacks
Regulatory challenges	• Varying global laws on data and UBI bans on data collection in certain states e.g. California currently has a ban on collecting Telematics data.	• Legal risks and could result in lawsuits over discriminatory premiums. • Surveillance can cause backlash, ethical debates and potential lawsuits rom customers.
Limited Adoption	• Not all vehicles are currently compatible - older models lack sensor.	• Drivers with older vehicles that lack sensors, will loose out on the incentives offered by the insurers.

Market Research

Demand

- Vehicle Maintenance and Operations team
- Data Analytics and IT Teams
- Manufacturer Relations/Notifications Team
- Insurers: Insurance and Risk Management Teams
- Safety and Compliance Team
- Customer Support and Fleet Management team

Competition

- Mid-Large sized auto manufacturers becoming digital natives and incorporating IoT sensors within their cars

Project Metrics

Budget

Development Costs:

- For the initial development of the PoC the cost can range from $20,000 to $50,000 covering AI modeling, IoT integration, and data analytic capabilities
- AI deployment in PoC costs can vary from $5,000 to $15,000 focusing on validating AI feasibility.
- IoT app development for the PoC might cost $10,000 to $20,000 limited to sensor integration on a few vehicles

Operational Costs:

- Monthly running costs can vary from $200 to $1000 monthly

Risk

- **Cybersecurity Threats:** The connected vehicle systems are vulnerable to hacking attacks such as manipulation of braking, steering, engine, or transmission control systems
- **Privacy and Data Protection:** Continuous GPS and telematics data collection
- **Connectivity and Reliability:** Dependence on network connectivity is critical for real time data transmission and predictive functionalities/ Loss of connectivity could lea to failures in navigation, vehicle operation, or timely intervention alerts, impacting safety.
- **AI Risks:** AI components could mis-detect objects or lanes, leading to incorrect vehicle behavior such as collisions or wrong path driving.

Timeline

Typical timeline for an integrated AI + IoT vehicle telematics platform can be delivered in 3 to 6 months

PoC Phases:

- Requirements definition and scoping (2-3 weeks)
- Hardware and telematics sensor setup (3-4 weeks)
- Development of data ingestion pipelines and initial AI models (4-6 weeks)
- Integration with manufacturer notification and insurance incentive system in simplified form (3-4 weeks)
- Testing, validation and demonstration of actionable insights (2-3 weeks)

Technology

To successfully pull off PoC for the integrated Ai+ IoT vehicle telematics platform the following technology stack should be used:

Hardware Components:

- Telematics Control Unit (TCU)
- GPS modules for location tracking
- Cellular modems (4G/5G) for secure, real-time data transmission to the cloud
- Sensors capturing the entire diagnostics, driven behavior, and environmental inputs

Software Components:

- IoT middleware for data ingestion, filtering and secure transmission
- Cloud platform (AWS, Azure, or Google Cloud) for scalable data storage, processing, and analytics
- AI/ML frameworks (TensorFlow, PyTorch) to build predictive maintenance models and behavior analysis
- API's and backend services to generate manufacturer notifications and insurance incentive triggers
- Dashboard or mobile apps for real time visualization and alerts

Communication Protocols:

- CAN, LIN for vehicle networking interfacing
- Secure MQTT or HTTPS protocols for IoT data communication
- Over-the-air update capability for firmware and AI model improvements

Resources

The following human resources will are essential:

- IoT hardware engineers to select and integrate telematics sensors and devices with the vehicles
- AI/ML data scientists and software developers to build predictive maintenance models and data analytics pipelines
- Cloud engineers for infrastructure setup on AWS, Azure or Google Cloud.
- Backend developers to create APIs, notification systems and integration with manufacturer and insurer systems
- UX/UI designers to develop dashboard or mobile apps for real time monitoring
- Cybersecurity specialists to secure data transmission and privacy compliance
- Project managers to coordinate activities and timelines

PoC Timeline

Requirements & Planning

2-3 weeks

Key Activities

- Define scope, data sources, KPI's and PoC goals

Hardware and Sensor Setup

3-4 Weeks

Key Activities

- Selection & Installation of telematics units and core sensors on test vehicles

MS1

Data Pipeline and Middleware

3-4 Weeks

Key Activities

- Develop IoT data ingestion, filtering, and secure communication layers

AI Model Development

4-6 Weeks

Key Activities

- Build and train predictive maintenance and drover behavior AI models with core datasets

MS2

Integration & Notification

3-4 Weeks

Key Activities

- Implement manufacturer notifications and insurance incentive logic in a simplified form

Testing and Validation

2-3 Weeks

Key Activities

- Functional, data accuracy, and initial security testing

MS3

Demonstration & Feedback

1-2 Weeks

Key Activities

- Present PoC results to stakeholders, gather feedback for full solution roadmap

MS4

Key:

MS1: Planning & Setup
MS2: AI Model Dev complete
MS3: Integration & Testing complete
MS4: Demo & Feedback

Validation Metrics/Success Criteria

Some suitable validation metrics and success criteria for this PoC of the integrated AI+ IoT vehicle telematics platform are:

1. **Data Accuracy and Completeness:** Measures the correctness and fullness of vehicle and sensor aew collected, ensuring minimal missing or erroneous data points
2. **Predictive Maintenance Accuracy:** Use KPIs like Mean Time Between Failures (MTBF), Mean Time to Repair (MTTR). Monitor both leading indicators (vibrations, temperature) and lagging indicators (downtime, repair costs) to validate predictive accuracy.
3. **Real-time Notification Latency:** Measure the delay from event detection (e.g. fault driver behavior anomaly) to notification delivery. Low latency ensures timely intervention.
4. **Connectivity Uptime:** Track the percentage of time the IoT network and devices maintain active connection and data transmission without interruptions, targeting high availability standards typical in IoT monitoring systems.
5. **Driver Behavior Improvement:** Evaluate changes in driver behavior over time through metrics such as reduction in harsh braking, acceleration, speeding incidents and adherence to safe driver stores.
6. **System Usability:** Collect user feedback and perform usability testing to verify ease of interaction with the telematics platform
7. **Security & Privacy Compliance**: Assess compliance with relevant data security standards e.g. GDPR, CCPA by auditing access controls , data encryption, and user content management.

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